algorithmica

Generative AI
(Technology & Applications/Agents)

Course Details

- **Language Choice:** **Python3**
- **Fee:** **25000/**
- **Session Planning:** **2 sessions/week**
- **Session Timings:**

  Saturday & Sunday – 7.30am to 9.30am (IST)
- **Duration:** **3 1/2 – 4 months**
- **Mode of Delivery:** **Online LIVE**

Course Curriculum

Part-I: Generative AI Agents/Applications

1. **Agent Basics**
   - What are agents?
   - Agent components: Foundational models, tools, memory, guardrails, etc.,
   - Async io vs multi-threading
   - Prompt engineering
   - Working with foundation models: Gemini, openai, huggingface, claude, etc.,
   - Technology stack
2. **Agent Architectures & Patterns**
   - Single vs Multi agent
   - Conversational vs Non-conversational
   - Streaming vs non-streaming

Copyright © Algorithmica
www.algorithmicaonline.com

- Delegation vs hand-off
- Memory types
- Reflection
- Guardrails
- Human-in-the loop
- RAG
- Model agnostic agents

3. **Evaluation of AI Systems**
   - Metrics
   - Strategy
4. **Protocols**
   - MCP
   - A2A
   - A2UI
5. **Text Search**
   - Traditional Search
   - Semantic Search
6. **Skills**
   - Question answering
   - Structure extraction
   - Language translation
   - Text Summarization
   - Text Classification
   - Tool calling
7. **Issues & Risks in AI systems**
   - Hallucination
   - Privacy
   - Security
   - Data protection

Copyright © Algorithmica
www.algorithmicaonline.com

- Compliance, Safety & Regulations
- Non-determinism

8. **Sample Agents**
   - Coding Agent
   - Enterprise data query agent
   - Database query agent
   - Document processing agent
   - Test Automation agent
   - Multimodal agents

Part-II: Generative AI Technology Creation

1. **Models of intelligence**
   - Symbolic models
   - Neural network models
   - Understanding AI, Machine learning, deep learning and generative learning
   - Technology stack
2. **Predictive NN models**
   - DNN Architectures
   - DNN Model Training, Deployment and Inference
   - DNN Training Algorithm & Math understanding
3. **Generative NN models**
   - GNN Architectures: Embedding layers, transformer layers
   - GNN Model Training, Deployment and Inference
   - GNN Training Algorithm

Copyright © Algorithmica
www.algorithmicaonline.com

Generative AI
(Technology & Applications/Agents)

4. **Large Language Models(LLM)**
   - LLM Architectures
   - LLM Training
   - Pretraining
   - Supervised Fine Tuning: LoRA, QLoRA
   - Preference Tuning
   - LLM Deployment and Inference
   - LLM Training Algorithm
   - LLM Evaluation
5. **Large Multimodel Models(LMM)**

Copyright © Algorithmica
www.algorithmicaonline.com